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(54) **FLUID MONITORING**

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(58) **Field of Classification Search**

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See application file for complete search history.

(56)

References Cited

U.S. PATENT DOCUMENTS

2017/0003285 A1* 1/2017 Sundvor G01N 33/5308
2019/0234463 A1* 8/2019 Lugt F16C 33/6633

FOREIGN PATENT DOCUMENTS

CN 108008119 A * 5/2018 G01N 33/2888
JP 2022052591 A * 4/2022
WO WO-2014167503 A1 * 10/2014 G01N 11/00
WO WO-2017007333 A1 * 1/2017 E21B 19/06

OTHER PUBLICATIONS

Rezasoltani, A. Khonsari, M. M. on the Correlation between Mechani-
cal Degradation of Lubricating Grease and Entropy, Tribology
Letters 60 (1), (2015): 1-4.

(Continued)

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ABSTRACT

Methods of evaluating mixing are disclosed relating to the
preparation and mixing of materials. Materials prepared may
transition from heterogeneous mixtures to homogenous mix-
tures while the current of a motor is monitored. Quantities
relating to the rate of change of power or the rate of change
of entropy may be calculated in evaluating the sufficiency of
mixing. Methods of evaluating lubricant degradation are
also disclosed.

20 Claims, 10 Drawing Sheets

